LUKE THOMPSON

UNREAL ENGINE 5 TOOLS & PIPELINE PROGRAMMER (PYTHON / C++)

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PROFESSIONAL SUMMARY

Tools-minded UE5 developer. Since February 2026 I've built editor tooling, asset validation, and build automation for HomeWorld, my solo Unreal Engine 5.8 / C++ project. I design each tool and the rules it enforces, and I direct and review AI-assisted implementation behind tests. Shipped Shift (Phaser 3 / TypeScript), and did contract game work in C# at OutFusion. Remote freelance from Canada, Eastern Time.

TOOLS & PIPELINE (HOMEWORLD, UE 5.8)

- **Asset validation on save:** five Python `EditorValidatorBase` validators in Epic's Data Validation plugin (naming prefixes per class, missing hard references and empty material slots, texture size and power-of-two, map ownership, no `/Game/Developers` references). Registered at editor start. The rules are pure Python, tested on the host without an editor, and also run as a whole-project batch pass.
- **Headless test runner that refuses false greens:** launches or attaches to the editor, runs Automation tests, and hard-fails on a UTF-8 BOM report, a wrong engine version, zero tests run, a stale report, or a DLL older than its source. Each had once produced a fake pass. Plus a PowerShell build wrapper with an editor-close, retry, and build-queue protocol.
- **Data-driven placement:** the metrics sheet is schema-checked before any actor moves, placers run in `ScopedEditorTransaction` (one Ctrl+Z reverts a whole run), and a metrics gate scores ground contact, float gap, bounds, and overlap into a pass, soft-fail, or closed-fail JSON verdict.
- **Blender greybox pipeline:** a spec reader that measures origin, bounds, pivot, poly count, and material binding against specs (read-only by default, never overwrites authored meshes), and a JSON recipe-to-mesh builder in headless Blender that emits `UCX_` collision and a sha256 checklist.
- **Editor automation:** a JSON request/response bridge for scripts and agents, a Blueprint-to-JSON inspector, and C++ commandlets in a `HomeWorldEditor` module (PCG graph setup, Mass Entity Config creation). **In progress:** a nightly BuildCookRun package with a boot smoke test and a one-line verdict.

CORE SKILLS

Languages	Python, C++, C#, PowerShell, TypeScript/JavaScript, SQL
UE5 tooling	Editor Python (`unreal`), Data Validation, Asset Registry, commandlets, editor transactions, Automation tests, UBT/BuildCookRun, PCG, Movie Render Queue
UE5 gameplay	Gameplay Ability System, custom CharacterMovementComponent modes, Enhanced Input, Blueprints
Pipeline	Blender Python (bpy), MCP editor bridges, JSON schemas, pytest, Git/GitHub, AI-assisted dev with review gates

SELECTED PROJECTS

HomeWorld — Unreal Engine 5.8 / C++ game Feb 2026 – present
Independent (solo) · UE 5.8, C++, Python · private repository, available on request

- Third-person ARPG survival and family-sim: a custom glide mode on the CharacterMovementComponent, a traversal component (mantle, vault, blink), six GAS abilities on a Blueprintable C++ base, and data-driven interactions and zones. Backed by 42 C++ automation test files and 150+ Python test files for the tooling.

Shift — playable shop & delivery game Aug – Sep 2026
Independent · Phaser 3, TypeScript, Vite · Live: xylardark.github.io/ShiftGame

- Shipped a browser/PWA game loop: fixed 60 Hz simulation, render interpolation, cached A* routing, and a tiered render budget for phones.

Portfolio Studio — media-résumé site with per-employer pages Sep 2026 – present
Independent · React, TypeScript, Supabase · xylardark.github.io/Portfolio-Studio

RELEVANT EXPERIENCE

Video Game Developer · Contract — OutFusion · Hamilton, ON 2022

- Integrated and tested game UI art assets across mobile and desktop. Built and tested C# REST API integrations for player data, and partnered on bug verification and release quality.

Software Developer · 12-Month Contract — TruckSail · Ancaster, ON 2020

- STM32/CAN firmware and defect finding, JS/Bootstrap UIs, SQL optimization, and GitHub workflows.

Delivery Driver (Uber, retail delivery) · Hamilton, ON 2022 – 2026

EDUCATION

Diploma, Computer Systems Technology — Software Development · Mohawk College · Hamilton, ON · 2019